RestoPulse — Master Documentation Index & Curriculum Guide

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Welcome to the definitive documentation index for **RestoPulse** (Cloud Restaurant POS & Kitchen ERP).

Use this guide to know **which document to read**, **what to explain to whom**, and **which manual covers which part of the system**.

Quick Reference: Which Document Do I Use For What?

Your Goal / Who You Are Talking To	Document to Use	Location
Explain Product Features & Day-to-Day Operations to a Restaurant Owner	`SPOTLIGHT_FULL_OPERATIONS_MANUAL.md`	`docs/SPOTLIGHT_FULL_OPERATIONS_MANUAL.md`
Explain Code Architecture Line-by-Line to a Developer or Technical Interviewer	`CODE_WALKTHROUGH_LINE_BY_LINE.md`	`docs/CODE_WALKTHROUGH_LINE_BY_LINE.md`
Explain Cloud Tech Stack, Database & Hosting Architecture	`ARCHITECTURE.md`	`docs/ARCHITECTURE.md`
How to Pitch, Overcome Objections & Close Sales Deals with Restaurant Owners	`RESTAURANT_SALES_BATTLECARD_AND_CLOSING_STRATEGY.md`	`docs/RESTAURANT_SALES_BATTLECARD_AND_CLOSING_STRATEGY.md`
Explain Restaurant Business Economics, Unit Economics & Margins	`BUSINESS-PLAYBOOK.md`	`docs/BUSINESS-PLAYBOOK.md`
How to Setup, Install, and Run Locally from Scratch	`SETUP.md`	`docs/SETUP.md`
Full Technical & Business Master Reference	`THE_DEFINITIVE_MASTER_ENGINEERING_AND_BUSINESS_BIBLE.md`	`docs/THE_DEFINITIVE_MASTER_ENGINEERING_AND_BUSINESS_BIBLE.md`

(Everything in One Place)		
Testing, Quality Assurance & Edge-Case Verification	`TESTING.md`	`docs/TESTING.md`
How to Customize Menu, Recipes, Prices & Tax Rates	`CUSTOMISE.md`	`docs/CUSTOMISE.md`

Detailed Breakdown of Each Document

1. `SPOTLIGHT_FULL_OPERATIONS_MANUAL.md`

- **Purpose:** Complete end-to-end operational handbook for restaurant owners, billing counter cashiers, and kitchen chefs.
- **What it covers:**
- **Billing Terminal Workflow:** How to take a counter order, apply discounts, select Cash/UPI/Card, and print 80mm/58mm thermal receipts in under 3 seconds.
- **Deg Batch Portions & Stock Cap:** Why authentic biriyani kitchens cook in batches (e.g. 160 plates per deg) and how the live countdown prevents selling over capacity.
- **WhatsApp Digital Receipts:** Sending direct bill receipts over WhatsApp without costly SMS gateway fees.
- **Raw Material Inventory:** Logging raw meat, basmati rice, ghee, and cooking oil with low-stock alerts.
- **Multi-Branch Operations:** Switching between Taramani Flagship, Adyar, and Velachery branches with separate stock and sales logs.
- **When to read it:** Read this before demonstrating the software to a restaurant manager or cashier staff.

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2. `CODE_WALKTHROUGH_LINE_BY_LINE.md`

- **Purpose:** Engineering deep-dive explaining the TypeScript and React code line by line.
- **What it covers:**
- **State Store (`src/lib/pos-store.ts`):** How `getDB()` and `setDB()` maintain reactive client state with zero external library overhead.
- **Bidirectional Cloud Sync (`src/lib/neon-sync.ts`):** How `createServerFn` executes server-side queries to Neon PostgreSQL via connection-pooled HTTP endpoints.
- **Reactive Rendering:** How `useMemo` optimizes search, dish category filtering, and sales calculations without UI lag.
- **Audio Synthesis:** How the HTML5 Web Audio API generates authentic cash-register chimes without external MP3 audio files.
- **When to read it:** Read this when explaining how the application is built to a software engineer, technical interviewer, or co-founder.

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3. `ARCHITECTURE.md`

- **Purpose:** High-level system design and cloud infrastructure document.
- **What it covers:**
- **Zero-Cost Cloud Deployment:** How the application runs 100% free on Vercel Edge Serverless and Neon Cloud PostgreSQL (AWS Singapore region).
- **Bidirectional Offline-First Sync:** Why the app works offline at the counter even if Wi-Fi temporarily drops, and syncs automatically when connection is restored.
- **Data Schema:** The exact PostgreSQL schema for `"Sale"` , `"SaleLine"` , `"DishItem"` , and `"Stock"` .
- **When to read it:** Read this when evaluating scaling, security, data persistence, and database backups.

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4. `RESTAURANT_SALES_BATTLECARD_AND_CLOSING_STRATEGY.md`

- **Purpose:** The commercial sales playbook for pitching, negotiating, and closing restaurant contracts.
- **What it covers:**

- **Why Restaurants Need This:** Eliminating staff pilferage, meat shrinkage, Swiggy/Zomato commission leakage (24%), and expensive hardware lock-ins.
- **Objection Handling Scripts:**
- *"I don't have a laptop"* → How to close using any Android mobile phone or budget tablet.
- *"I already use Petpooja / Posist"* → Feature-by-feature comparison showing cost savings and biriyani-specific batch tracking.
- *"Will my financial data be safe?"* → How enterprise cloud encryption and Neon PostgreSQL protect their sales records.
- **The 2-Minute Walk-In Demo Script:** Exact words to say when meeting a restaurant owner in person.
- **When to read it:** Read this every time before speaking with a prospective client or restaurant owner.

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5. `BUSINESS-PLAYBOOK.md`

- **Purpose:** Restaurant economics and financial modeling.
- **What it covers:**
- Food cost percentage breakdown (meat 32%, rice 8%, spices 5%).
- Swiggy vs Zomato vs Counter Dine-in margin calculations.
- Breakeven plate calculations for cloud kitchens and quick-service restaurants (QSR).
- **When to read it:** Read this to speak the financial language of restaurant owners (gross margin, net margin, wastage percentage).

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6. `SETUP.md` & `TESTING.md`

- **Purpose:** Developer onboarding, local environment setup, and test suite execution.
- **What it covers:**
- Node.js version requirements, npm install steps, and environment variables (`DATABASE_URL`).
- Unit tests, edge cases, and thermal printer simulation tests.
- **When to read it:** Read when installing the project on a new computer or preparing a fresh deployment.

Study Roadmap: What Order Should You Read?

1. **Day 1 — Master the Product:** Read `docs/SPOTLIGHT_FULL_OPERATIONS_MANUAL.md` to know every single button and feature.

2. **Day 2 — Master the Sales Pitch:** Read `docs/RESTAURANT_SALES_BATTLECARD_AND_CLOSING_STRATEGY.md` to know how to sell and overcome any doubt.

3. **Day 3 — Master the Tech & Code:** Read `docs/ARCHITECTURE.md` and `docs/CODE_WALKTHROUGH_LINE_BY_LINE.md` to understand the code under the hood.